

**A REPORT
ON
RAPIDO RIDE ANALYTICS DASHBOARD – BANGALORE**

By

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1. Introduction

With the rapid growth of app-based transportation services, large volumes of ride-related data are generated daily. Analyzing this data is essential for understanding operational performance, customer behavior, and service efficiency. This project focuses on analyzing ride data from Rapido services in Bangalore and presenting the insights through an interactive and visually structured dashboard using Microsoft Excel. The project demonstrates how raw transactional data can be transformed into meaningful information using data cleaning, aggregation techniques, and visualization tools. The final dashboard provides a consolidated view of key metrics such as total rides, revenue, cancellations, and usage patterns, enabling easy interpretation and decision-making.

2. Objectives

The main objectives of this project are:

- To analyze ride data and identify trends in demand over time
- To calculate important business metrics such as total rides, revenue, and average fare
- To examine ride completion and cancellation patterns
- To study customer payment preferences and service usage
- To design a dynamic and interactive dashboard using Excel features

3. Dataset Description

The dataset used in this project contains detailed ride-level information collected from Rapido services in Bangalore. Each record represents a single ride and includes the following fields:

- **Service_Type:** Indicates the type of service (Auto, Bike, Bike Lite, Cab Economy, Parcel)
- **Ride_Date and Ride_Time:** Represents when the ride was booked or completed
- **Ride_Status:** Specifies whether the ride was completed or cancelled

- **Pickup_Location and Drop_Location:** Shows the origin and destination of the ride
- **Duration_Min:** Time taken to complete the ride
- **Distance_KM:** Distance travelled during the ride
- **Base_Fare, Extra_Charges, Total_Fare:** Financial details of the ride
- **Payment_Method:** Mode of payment used by the customer

This dataset provides sufficient information to analyze both operational and financial aspects of ride services.

4. Data Cleaning and Preparation

Before analysis, several preprocessing steps were performed to ensure data quality and consistency:

- Missing or invalid values such as “nan” in payment fields were identified and handled
- Date and time columns were formatted properly to ensure accurate grouping and analysis
- Numeric columns such as fare, distance, and duration were verified and converted to appropriate formats
- Duplicate or inconsistent categorical entries were checked and standardized

These steps were essential to avoid errors during pivot table creation and ensure accurate results.

5. Data Analysis Using Pivot Tables

Pivot tables were the primary tool used for summarizing and analyzing the dataset. Multiple pivot tables were created to extract insights from different perspectives.

For example:

- A pivot table was created with **Count of Ride_ID** to calculate total rides
- Another pivot table used **Sum of Total_Fare** to calculate total revenue
- The **Average of Total_Fare** was used to determine average fare per ride
- A pivot table with **Ride_Status** was used to calculate cancellation percentage

Additionally, pivot tables were used to group data by:

- Month (to analyze ride trends over time)
- Service type (to compare revenue contribution)
- Hour of the day (to analyze ride demand patterns)
- Payment method (to study customer preferences)

These pivot tables formed the foundation for both KPI calculations and visualizations.

6. Key Performance Indicators (KPIs)

Based on pivot table analysis, the following KPIs were identified and displayed on the dashboard:

- **Total Rides:** Represents the total number of rides recorded in the dataset
- **Total Revenue:** Total earnings generated from all rides
- **Cancellation Rate:** Percentage of rides that were cancelled
- **Average Fare:** Average cost per ride

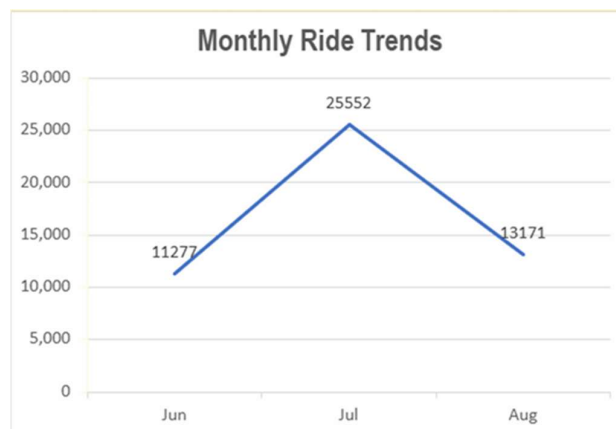
For instance, cancellation rate was calculated by dividing the number of cancelled rides by total rides and converting it into percentage format.

These KPIs were linked directly to pivot tables so that they update automatically when filters are applied.

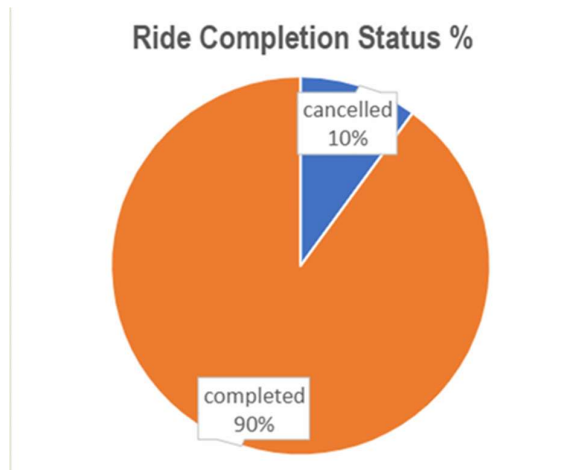
7. Data Visualization

To make the analysis more intuitive, various charts were created:

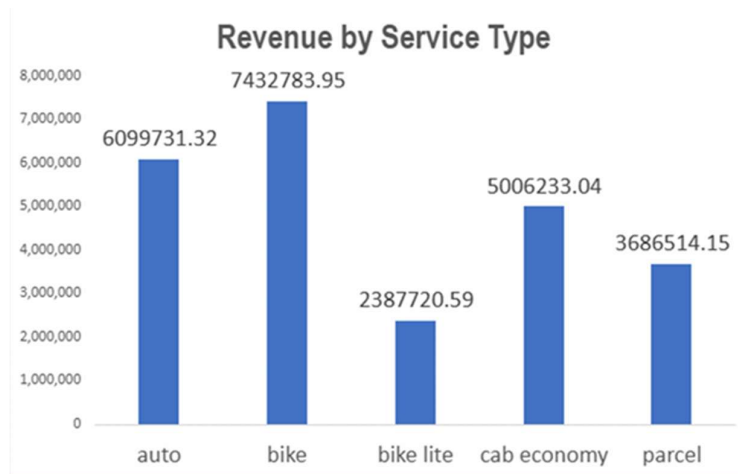
- A **line chart** was used to display ride trends over different months, helping identify peak periods



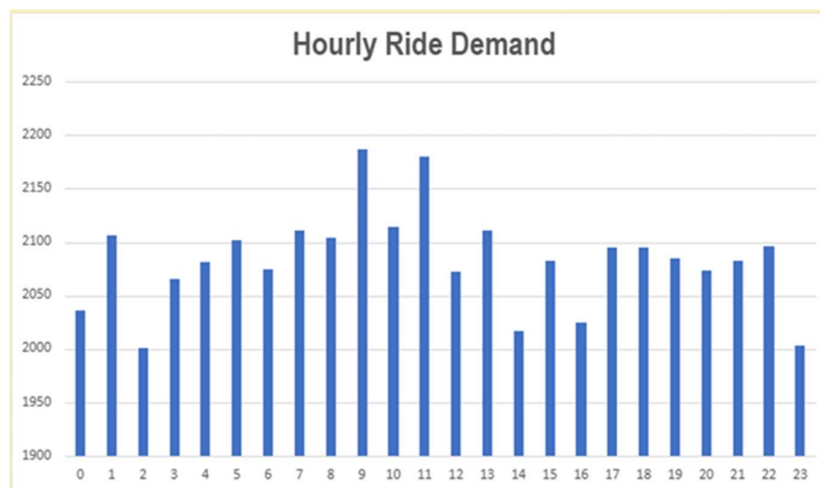
- A **pie chart** was used to represent the proportion of completed and cancelled rides



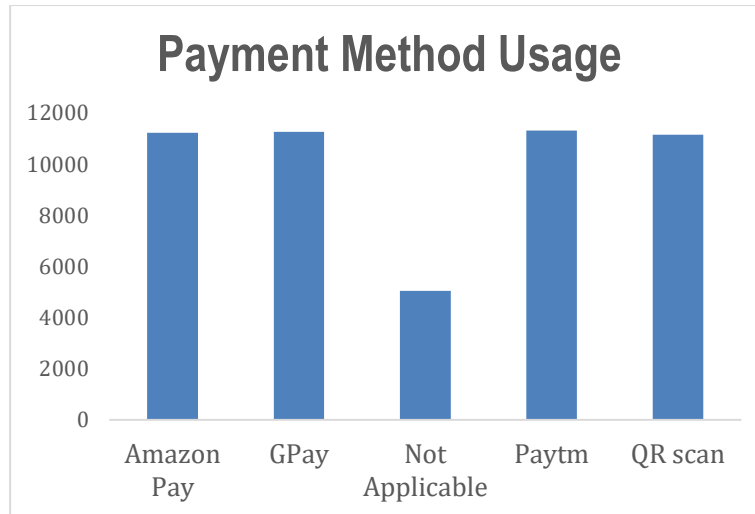
- A **column chart** was used to compare revenue generated by different service types



- A **bar chart** was used to analyze ride demand across different hours of the day

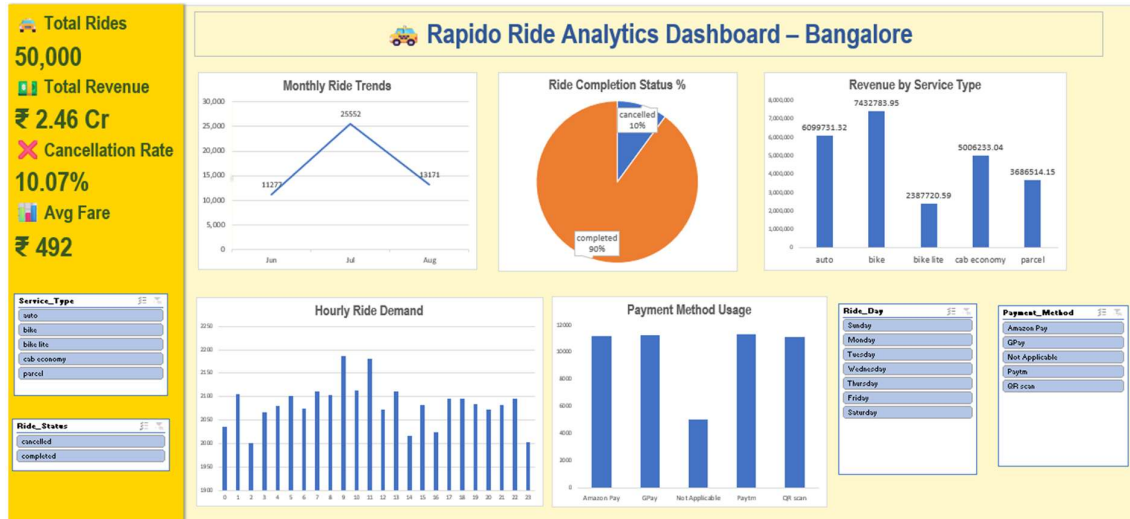


- Another **column chart** was used to display distribution of payment methods



Each chart was designed to highlight a specific aspect of the dataset and support easy comparison.

8. Results



9. Dashboard Design and Layout

The dashboard was designed with a clear and structured layout to ensure readability and visual appeal:

- A vertical panel on the left side was used to display KPI cards
- The main area was divided into sections for charts, arranged systematically

- A light yellow background theme was used to align with Rapido branding
 - Consistent font styles, sizes, and spacing were applied across the dashboard
 - Shapes and text boxes were used to present KPI values clearly and prominently
- Special attention was given to alignment, spacing, and formatting to create a clean and professional appearance.

10. Interactivity Using Slicers

To enhance usability, slicers were added to enable interactive filtering. The slicers included:

- Service Type
- Ride Status
- Ride Day
- Payment Method

These slicers were connected to all relevant pivot tables and charts using the report connections feature. As a result:

- Selecting a specific service type updates all charts and KPIs accordingly
- Filtering by ride status shows only completed or cancelled rides
- Selecting a particular day allows analysis of daily patterns

This interactivity allows users to explore the data dynamically without modifying the underlying dataset.

11. Key Insights

The analysis of the dataset led to several important observations:

- Ride demand showed variation across months, with one month having significantly higher activity
- A majority of rides were successfully completed, indicating efficient service operations
- Certain service types contributed more to revenue, highlighting their importance in business performance
- Digital payment methods were widely used, indicating customer preference for cashless transactions

- Ride demand varied throughout the day, with certain time periods showing higher activity

These insights demonstrate how data analysis can support operational understanding and decision-making.

12. Challenges Faced

During the development of the project, several challenges were encountered:

- Difficulty in linking pivot table values dynamically to dashboard elements
- Errors while using formulas such as GETPIVOTDATA
- Formatting numerical values into Indian currency format (lakhs and crores)
- Managing multiple pivot tables and ensuring they respond correctly to slicers
- Designing a visually balanced and professional dashboard layout

These challenges were resolved through step-by-step debugging, formatting adjustments, and repeated testing.

13. Conclusion

This project successfully demonstrates the use of Microsoft Excel as a powerful tool for data analysis and dashboard creation. By applying data cleaning techniques, pivot table analysis, and visualization methods, a comprehensive and interactive dashboard was developed.

The dashboard provides a clear overview of ride performance, revenue trends, and customer behavior, making it a useful tool for analysis and reporting. The use of slicers adds interactivity, allowing users to explore the data from different perspectives efficiently.